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[PROGRAMMING 2B]

[RACEDAY ASSIGNMENT PART 1]

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RaceDay API Endpoint Plan

1. Introduction

The Raceday system will use a REST API to manage users, participant profiles, event types, events, categories, enrolments and race results.

The API is planned for two main types of users: **Organisers** and **Participants**. Some information, such as available events and published results, can also be viewed by members of the public.

This endpoint plan describes how the API will work before development starts. Each endpoint includes the HTTP method, route, purpose, required role, request body and expected response.

The API will be developed in Part 2. The design below is based on the RaceDay database and its relationships from Part 1.

2. User Roles

Organiser

An Organiser creates and manages Raceday events. They can create event types and categories, view participants enrolled in their events, manage enrolment statuses and record race results.

Participant

A Participant can create a participant profile, view available events, select an event category, enrol in an event and view their own enrolments and results.

Public User

A public user does not need to be logged in to register, log in, view available events and view results that have been made available for public viewing.

3. Authentication

Authentication comes first. Users need an account before they can access features that require a specific role.

Method	Route	Description	Role Required	Request Body	Expected Response
POST	`/api/auth/register`	Creates a new RaceDay account. The user can register as an Organiser or Participant.	Public	FirstName, LastName, Email, Password, Role, PhoneNumber	**201 Created** – Account created successfully with user information.
POST	`/api/auth/login`	Checks the user's login details and authenticates the account.	Public	Email, Password	**200 OK** – Authentication token/session and basic user information.
POST	`/api/auth/logout`	Ends the current authenticated session.	Authenticated User	None	**200 OK** – User logged out successfully.

4. User and Participant Profile

A user's account contains their basic account information. Participants also have a separate profile containing information needed when entering a RaceDay event.

Method	Route	Description	Role Required	Request Body	Expected Response
GET	`/api/users/{id}`	Retrieves the details of a specific user.	Authenticated User	None	**200 OK** – User information is returned.

PUT	`/api/users/{id}`	Updates the user's account details.	Account Owner	FirstName, LastName, Email, PhoneNumber	**200 OK** – Updated user information.
GET	`/api/participant-profiles/{userId}`	Retrieves the participant profile linked to a user account.	Participant	None	**200 OK** – Participant profile is returned.
POST	`/api/participant-profiles`	Creates a participant profile for a registered participant.	Participant	UserID, DateOfBirth, Gender, EmergencyContactName, EmergencyContactPhone, ProfilePictureURL	**201 Created** – Participant profile created successfully.
PUT	`/api/participant-profiles/{id}`	Updates an existing participant profile.	Participant	DateOfBirth, Gender, EmergencyContactName, EmergencyContactPhone, ProfilePictureURL	**200 OK** – Updated participant profile.

5. Event Types

Raceday supports different types of events, such as running, walking and cycling. Event types are stored separately so that an event can be associated with an existing type.

Method	Route	Description	Role Required	Request Body	Expected Response
GET	`/api/event-types`	Returns all event types available in the system.	Authenticated User	None	**200 OK** – List of event types.
GET	`/api/event-types/{id}`	Returns information about one event type.	Authenticated User	None	**200 OK** – Event type details.

POST	`/api/event-types`	Creates a new event type.	Organiser	TypeName, Description	**201 Created** – New event type created.
PUT	`/api/event-types/{id}`	Changes the name or description of an existing event type.	Organiser	TypeName, Description	**200 OK** – Updated event type.
DELETE	`/api/event-types/{id}`	Removes an event type when it is no longer required.	Organiser	None	**204 No Content** – Event type deleted successfully.

6. Events

Events are the main part of the RaceDay system. Organisers create them, while participants can view the information before deciding whether to enrol.

Method	Route	Description	Role Required	Request Body	Expected Response
GET	`/api/events`	Returns a list of available Raceday events. Search and filtering can be supported through query parameters.	Public	Optional search/filter parameters	**200 OK** – List of available events.
GET	`/api/events/{id}`	Returns the full details of a selected event.	Public	None	**200 OK** – Event information.
POST	`/api/events`	Creates a new	Organiser	OrganiserID, EventTypeID,	**201 Created** –

		Raceday event.		EventName, Description, EventDate, Location, Distance, BannerImage URL, Status	Event created successfully.
PUT	`/api/events/{id}`	Updates information for an existing event.	Organiser	EventTypeID, EventName, Description, EventDate, Location, Distance, BannerImage URL, Status	**200 OK** – Updated event information.
DELETE	`/api/events/{id}`	Deletes an event from the system.	Organiser	None	**204 No Content** – Event deleted successfully.
GET	`/api/users/{userId}/events`	Shows all events created by a particular Organiser.	Organiser	None	**200 OK** – List of the Organiser's events.

7. Categories

Categories describe the different participation options available within an event. For example, an organiser could offer a 5 km, 10 km or 21 km category.

Method	Route	Description	Role Required	Request Body	Expected Response
GET	`/api/categories`	Returns all race categories stored in the system.	Authenticated User	None	**200 OK** – List of categories.

GET	`/api/categories/{id}`	Returns information about one category.	Authenticated User	None	**200 OK** – Category details.
POST	`/api/categories`	Creates a new category for an event.	Organiser	EventID, CategoryName, MinimumAge, MaximumAge, Distance, EntryFee	**201 Created** – Category created successfully.
PUT	`/api/categories/{id}`	Updates an existing race category.	Organiser	CategoryName, MinimumAge, MaximumAge, Distance, EntryFee	**200 OK** – Updated category information.
DELETE	`/api/categories/{id}`	Removes a category that is no longer available.	Organiser	None	**204 No Content** – Category deleted successfully.

8. Enrolments

The enrolment process connects a participant with an event and the category they have selected.

A participant must select an available event and category before an enrolment can be created.

Method	Route	Description	Role Required	Request Body	Expected Response
POST	`/api/enrolments`	Creates an enrolment for a participant in a	Participant	ParticipantID, EventID, CategoryID	**201 Created** – Enrolment created

		selected event category.			successfully.
GET	`/api/enrolments/my`	Returns all enrolments belonging to the currently logged-in participant.	Participant	None	**200 OK** – List of the participant's enrolments.
GET	`/api/events/{eventId}/enrolments`	Returns participants enrolled in a particular event.	Organiser	None	**200 OK** – List of event enrolments.
PUT	`/api/enrolments/{id}/status`	Changes the status of an enrolment, such as Pending, Confirmed or Cancelled.	Organiser	Status	**200 OK** – Updated enrolment status.

9. Results

Results are linked to enrolments. A participant may be enrolled before the event takes place, which means a result doesn't have to exist immediately.

Once the event has taken place, an Organiser can record the participant's finishing time and position.

Method	Route	Description	Role Required	Request Body	Expected Response

POST	`/api/results`	Records the result of a participant who completed an event.	Organiser	Enrolment ID, FinishTime, FinishPosition	**201 Created** – Result recorded successfully.
PUT	`/api/results/{id}`	Corrects or updates an existing race result.	Organiser	FinishTime, FinishPosition	**200 OK** – Updated result information.
GET	`/api/events/{eventId}/results`	Returns the results recorded for a particular event.	Public	None	**200 OK** – List of event results.
GET	`/api/results/my`	Returns the results belonging to the logged-in participant.	Participant	None	**200 OK** – Participant's results.
GET	`/api/results/{id}`	Returns one specific race result.	Authenticated User	None	**200 OK** – Result details.

10. Database Relationship and API Design

The API follows the relationships defined in the RaceDay database.

The **User** entity is connected to the **Participant Profile** entity. A user can have zero or one participant profile.

```
USER 1 ———————— 0..1 PARTICIPANT PROFILE
```

A user acting as an Organiser can create multiple events.

```
USER 1 ———————— M EVENT
```

Each event belongs to an event type. One event type can be used by multiple events.

```
EVENT TYPE 1 ———————— M EVENT
```

An event can contain multiple categories.

```
EVENT 1 ———————— M CATEGORY
```

A participant can enrol in multiple events.

```
PARTICIPANT PROFILE 1 ———————— M ENROLMENT
```

An event can have many enrolments because many participants can enter the same event.

```
EVENT 1 ———————— M ENROLMENT
```

A category can be selected by multiple enrolments.

```
CATEGORY 1 ———————— M ENROLMENT
```

Finally, an enrolment can have no result before the event takes place, or one result after the participant completes the event.

```
ENROLMENT 1 ———————— 0..1 RESULT
```

These relationships allow the API to move logically from registration, through event entry, and finally to race results.

11. Typical RaceDay Process

A normal user journey through the system could look like this:

```
User registers
      ↓
User logs in
      ↓
Participant creates profile
```

```
      ↓
Organiser creates event type
      ↓
Organiser creates event
      ↓
Organiser creates event categories
      ↓
Participant views available events
      ↓
Participant selects an event and category
      ↓
Participant creates enrolment
      ↓
Organiser views enrolments
      ↓
Participant takes part in the event
      ↓
Organiser records result
      ↓
Participant views result
```

This gives the API a clear sequence to follow. Each stage uses information created by the previous stage.

12. Access Control

Not every user should be allowed to perform every action.

A Participant shouldn't be able to create or delete another Organiser's event. In the same way, an Organiser shouldn't be able to change a participant's personal profile without the appropriate permission.

The API will therefore check the authenticated user's role before allowing restricted operations.

Public access

Public users can:

- Register an account.

- Log in.
- View available events.
- View published event results.

Participant access

Participants can:

- View and update their own account information.
- Create and update their participant profile.
- View available events.
- View event categories.
- Create an enrolment.
- View their own enrolments.
- View their own results.

Organiser access

Organisers can:

- Create and manage events.
- Create and manage event types.
- Create and manage categories.
- View enrolments for their events.
- Update enrolment statuses.
- Record race results.
- Update race results.

13. HTTP Status Codes

The API will return suitable HTTP status codes so that the client application can understand what happened after each request.

Status Code	Meaning
200 OK	The request was completed successfully.
201 Created	A new record was successfully created.
204 No Content	The requested resource was successfully deleted or no response body is required.
400 Bad Request	The information supplied in the request is invalid or incomplete.
401 Unauthorized	The user has not authenticated or supplied valid credentials.

403 Forbidden	The user is authenticated but doesn't have permission to perform the requested action.
404 Not Found	The requested record could not be found.
409 Conflict	The request conflicts with existing data, such as an enrolment that already exists.
500 Internal Server Error	An unexpected error occurred on the server.

14. Security

The API will require authentication for operations that involve personal information or restricted system functions.

Passwords will not be stored as plain text. They will be hashed before being stored in the database.

Role-based authorisation will be used to restrict Organiser and Participant functions. The API will also check that users can only update records they are permitted to change.

For example, a Participant can update their own participant profile, while an Organiser can manage events that they are responsible for.

15. Validation

The API will validate information before adding or changing records in the database.

Examples include checking that:

- Required fields have been supplied.
- Email addresses are valid.
- A selected event exists.
- A selected category exists.
- The category belongs to the selected event.
- A participant profile exists before creating an enrolment.
- The same participant cannot create duplicate enrolments for the same event and category.
- Event dates and other required values are valid.
- An enrolment exists before a result is recorded.

If validation fails, the API will return an appropriate error response rather than saving invalid information.

16. Endpoint Summary

The plan contains **33 endpoints** covering the main functions of the RaceDay system.

Area	Number of Endpoints
Authentication	3
User and Participant Profile	5
Event Types	5
Events	6
Categories	5
Enrolments	4
Results	5
Total	**33**

The number of endpoints is not intended to represent the number of database tables. The endpoints provide access to the functions of the system and allow the different entities to be created, viewed and managed.

17. Conclusion

This API Endpoint Plan provides the proposed structure for the RaceDay REST API before development begins.

The plan covers authentication, users, participant profiles, event types, events, categories, enrolments and results. It also defines which users can perform each operation and what information should be sent to and returned by the API.

The API will be implemented in Part 2 using the database structure designed in Part 1. The endpoint design will help keep the implementation organised and ensure that the API follows the relationships already defined in the RaceDay ERD.